CH32M030_PD Evaluation Board Reference

Version: V1.1

https://wch-ic.com

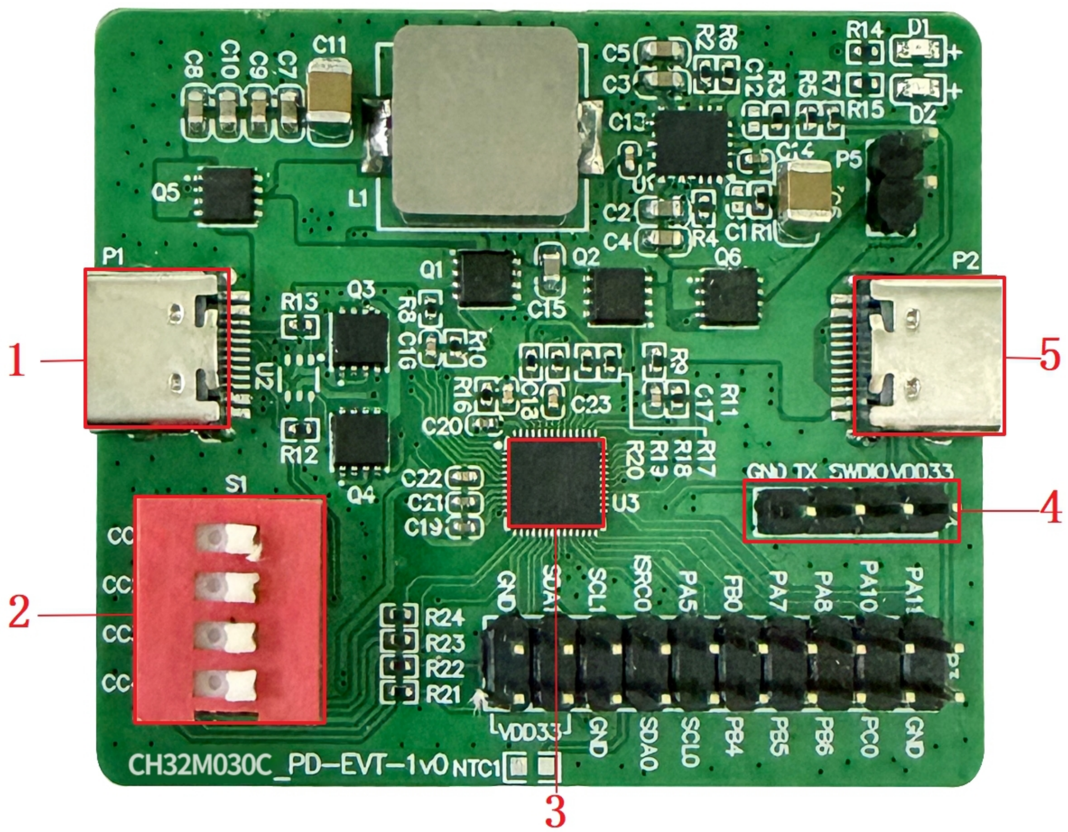
1. Overview

This evaluation board is only suitable for demonstrating the PD application routine under the EVT path (EVT->EXAM->APPLICATION->PD). The main control chip is CH32M030C9U7.

2. Evaluation Board Hardware

Please refer to the CH32M030SCH_PD.pdf document for the schematic of the evaluation board.

CH32M030 Evaluation Board



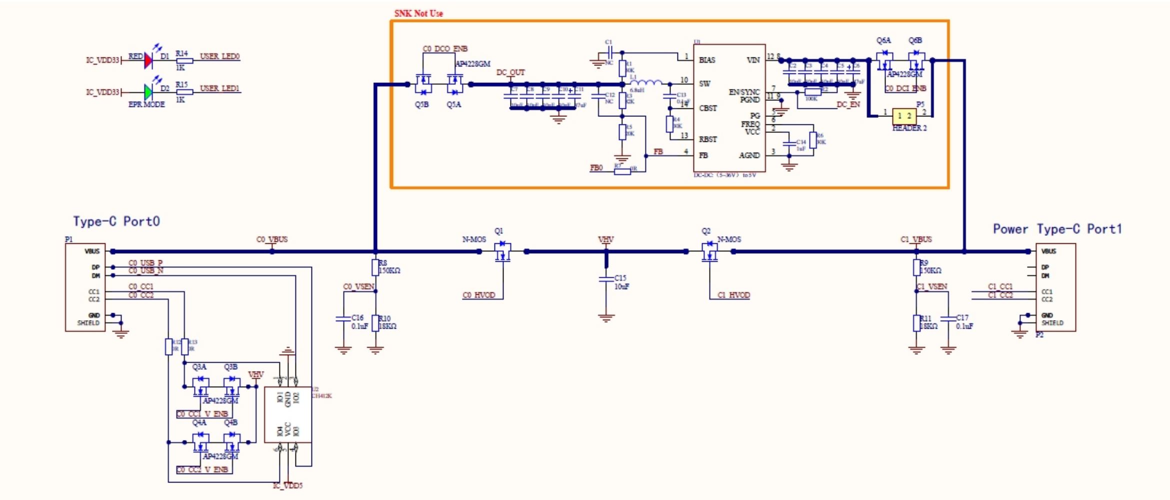
Descriptions

1. port0	2. CC port pull-down switch	3. MCU
4. SWD interface	5. port1	

2.1 The Above Evaluation Board Consists of the Following Modules

2.1.1 Power Module

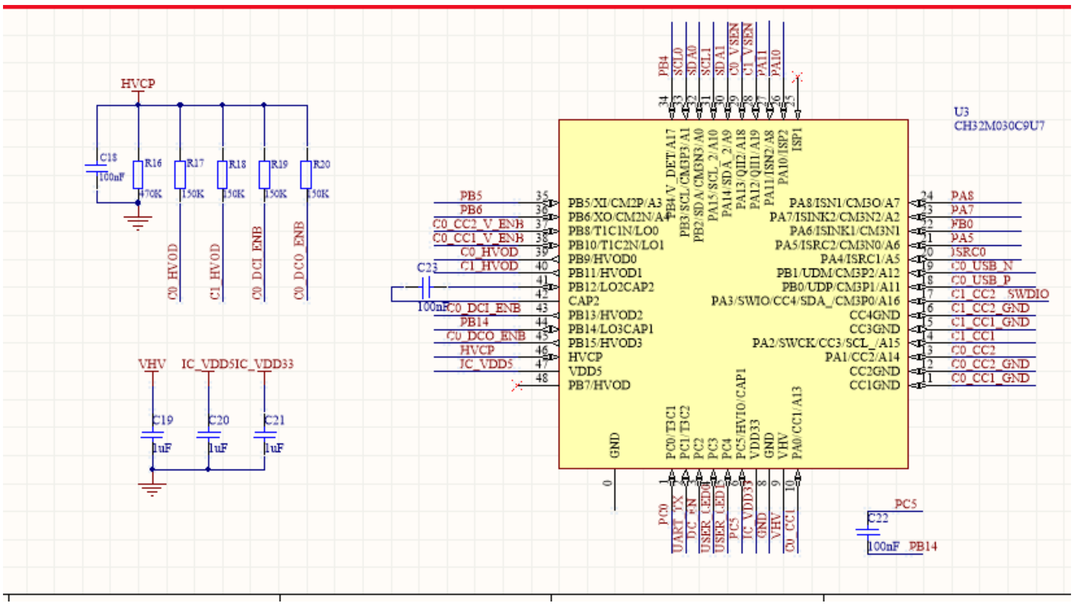
Figure 1 Power input interface



As shown in Figure 1, both TYPE-C interfaces can supply power to the MCU. P1 serves as a USB PD communication port. When acting as a SNK, the P1 port can induce different VBUS voltages, which, after passing through Q1, supply power to VHV. When acting as an SRC, P1 functions as an external power supply port. The P2 port directly induces voltages of the same level through handshake, which, after passing through Q2, supply power to VHV for the MCU. The MCU controls the power supply to P1 after enabling Q1. Alternatively, the P2 port induces a higher voltage, which, after passing through Q2, supplies power to VHV for the MCU. The other path is to step down the voltage through a DCDC (U1) and supply it to the P1 port, completing the power supply output.

2.1.2 MCU Circuit

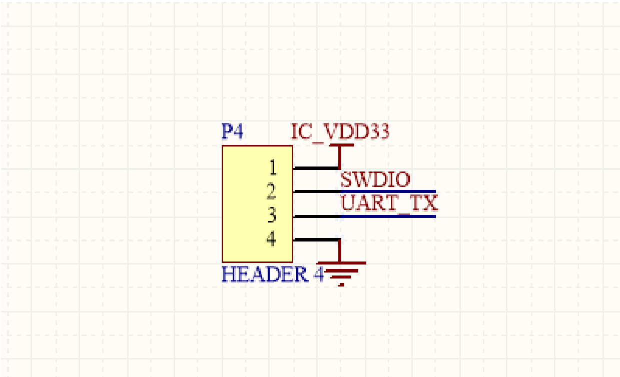
Figure 2 MCU circuit



As shown in Figure 2, the CH32M030C9U7 chip features an integrated high-voltage LDO, low static power consumption, high integration, and a minimal peripheral circuit. It also includes an internal N-type MOSFET gate boost driver module, capable of driving multiple high-side NMOSs for power path switching control.

2.1.3 SWD Interface

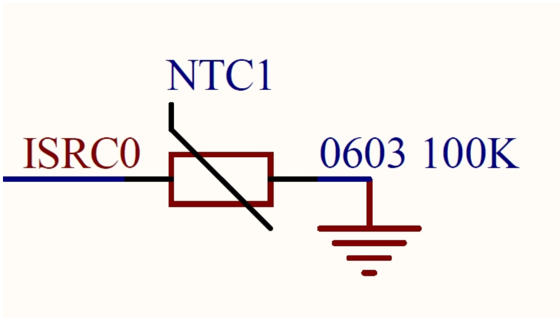
Figure 3 SWD interface



As shown in Figure 3, the SWD interface can be used for simulation debugging and downloading. This MCU model supports single-wire simulation download

2.1.4 ADC sampling temperature measurement circuit

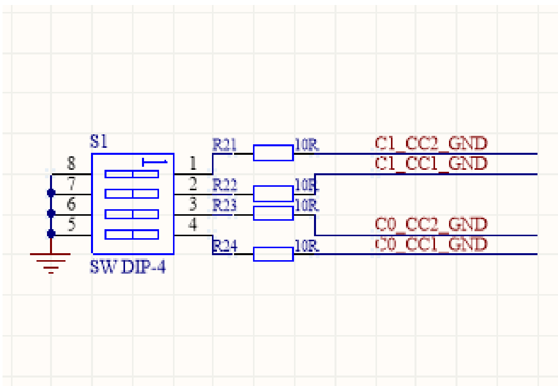
Figure 4 ADC sampling circuit



As shown in Figure 4, connect the thermistor to the ISRC0 pin, sample the voltage through ADC, and compare it to obtain the current temperature.

2.1.5 CC pull-down resistor selection circuit

Figure 5 CC pull-down circuit



As shown in Figure 5, when CC1GND, CC2GND, CC3GND, and CC4GND are grounded, the default Rd pull-down resistors of CC1, CC2, CC3, and CC4 are respectively activated. When they are left floating, they are not enabled. The selection of whether to connect the pull-down resistor is made through a dip switch, allowing for flexible conversion between the SNK and SRC roles of the PD.